

DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

MATHEMATICS – CLASS 10 UNIT ASSESSMENT

SUBJECT: MATHEMATICS

TIME ALLOWED: 2 HOURS

MAX MARKS: 41

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Write your name, roll number, and section clearly in the provided space.
2. All questions are compulsory.
3. Use of calculators is permitted where necessary.
4. Review all answers before final submission.

SECTION A - SHORT QUESTIONS

Attempt all questions. Each question carries 2 mark(s).

- Q1. State Ohm's Law. Mention the conditions under which it holds true. (2)
- Q2. Define electric potential and potential difference. Write their SI units. (2)
- Q3. Why are electrical heating elements (like in bread toasters or hair dryers) made of an alloy rather than a pure metal? (2)

SECTION B - NUMERICAL PROBLEMS

Attempt all questions. Each question carries 3 mark(s).

- Q1. An electric bulb is connected to a 220V generator. If the current drawn by the bulb is 0.50A, calculate the power of the bulb and its resistance. (3)
- Q2. An electric iron consumes energy at a rate of 840W when heating is at the maximum rate. The voltage is 220V. Calculate the current and the resistance. (3)
- Q3. A 100W electric bulb is lighted for 8 hours daily. Calculate the energy consumed in kilowatt-hours (kWh) by the bulb in a month of 30 days. (3)
- Q4. Find the equivalent resistance when two resistors of 10Ω and 40Ω are connected in parallel, and this combination is further connected in series with a 12Ω resistor. (3)
- Q5. A charge of 120 Coulombs flows through a bulb in 2 minutes. Find the magnitude of electric current passing through the circuit. (3)

SECTION C - LONG QUESTIONS

Attempt all questions. Each question carries 5 mark(s).

- Q1. Explain the fundamental concepts behind this topic, giving real-world examples and practical applications. (5)

- Q2. Briefly define the core terms associated with this subject matter and discuss their relevance. **(5)**
- Q3. Discuss the historical context and major scientific developments leading to our current understanding of this topic. **(5)**
- Q4. Explain the fundamental concepts behind this topic, giving real-world examples and practical applications. **(5)**

ANSWER KEY

SECTION A - SHORT QUESTIONS

Q1.	Answer not provided.	[2 marks]
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Q2.	Answer not provided.	[2 marks]
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Q3.	Answer not provided.	[2 marks]
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SECTION B - NUMERICAL PROBLEMS

Q4.	Answer not provided.	[3 marks]
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Q5.	Answer not provided.	[3 marks]
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Q6.	Answer not provided.	[3 marks]
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Q7.	Answer not provided.	[3 marks]
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Q8.	Answer not provided.	[3 marks]
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SECTION C - LONG QUESTIONS

Q9.	Answer not provided.	[5 marks]
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Q10.	Answer not provided.	[5 marks]
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Q11.	Answer not provided.	[5 marks]
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Q12.	Answer not provided.	[5 marks]
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